

Lab 3: Architectural Pattern Selection & Justification

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Assigned System: Remote Patient Vitals Alert & Monitoring App

Architecture Selection: "We chose Microservices Architecture for the Remote Patient Vitals Alert & Monitoring System."

1. Architectural Choice

We selected **Microservices Architecture** for the Remote Patient Vitals Alert & Monitoring System to decompose the solution into small, independently deployable services. This choice decouples high-throughput real-time telemetry stream ingestion from safety-critical anomaly detection, emergency alert escalation, and historical analytics rendering.

2. Two Scenario-Related Reasons

Reason 1: High-Concurrency Real-Time Telemetry Ingestion (FR-002, NFR-001): The system must continuously ingest real-time vital streams (SpO2, Heart Rate, Blood Pressure) from at least **500 connected remote monitoring devices** every 5 seconds with 99.99% uptime (<1% packet drop, <10s latency). Microservices architecture allows horizontal scaling of the Telemetry Ingestion Gateway independently to handle high peak data streams without choking downstream services.

Reason 2: Fault Isolation for Safety-Critical Alert Escalation (FR-001, FR-003, UC-04): Evaluating threshold breaches (HR > 140 BPM, SpO2 < 90%) within 2 seconds and escalating emergency alerts through a 60-second caregiver matrix are critical patient safety operations. Decoupling the Clinical Anomaly Evaluator and Alert & Escalation Engine into isolated services guarantees that heavy query loads from caregivers viewing 24-hour historical logs (FR-005) will never delay emergency alert dispatching.

3. Security Advantage (NFR-002 & HIPAA Compliance)

Per **NFR-002**, the architecture addresses security and regulatory concerns by enforcing **TLS 1.3**

encryption in transit for all REST, WebSocket, and MQTT payloads across public edge boundaries. Furthermore, patient health data is encrypted at rest using **AES-256** within the Encrypted Telemetry Store volume, maintaining strict **HIPAA compliance** and cryptographic isolation between public gateways and internal storage.

4. Performance Benefit (FR-001, FR-005)

By utilizing lightweight, asynchronous inter-service communications (gRPC stream buffers and priority event queues), incoming vital metrics are evaluated against clinical thresholds in **< 2 seconds** (FR-001). Pre-indexed timeseries logs allow the Vitals & Trend Analytics Service to render interactive 24-hour vitals trend graphs on caregiver dashboards in **< 2 seconds** (FR-005) even under peak system load.